

Product Carbon Footprint Study Report

Product Name: light emitting diode

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: light emitting diode

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1 kg light emitting diode

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1 kg light emitting diode.

2. System Boundary

light emitting diode production in China

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of light emitting diode produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **236.224 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

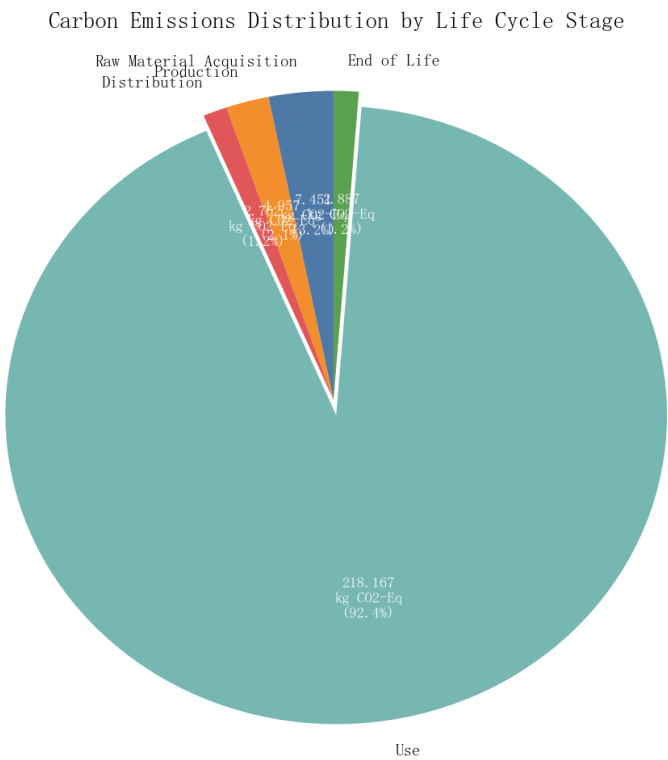


Table 1: Carbon Emissions by Life Cycle Stage

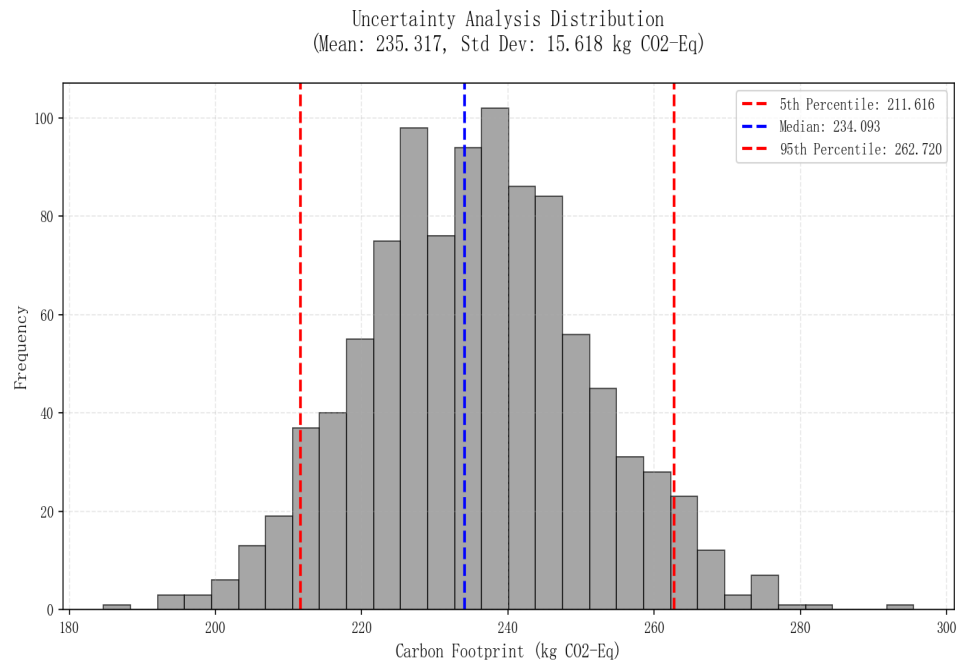
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	7.451	3.15
Production	4.957	2.10
Distribution	2.762	1.17

Use	218.167	92.36
End of Life	2.887	1.22
Total	236.224	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	235.317
Standard Deviation	15.618
Coefficient of Variation (CV)	6.64%
Median	234.093
5th Percentile	211.616
95th Percentile	262.720

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the light emitting diode is 236.224 kg CO₂-Eq, representing a moderate to high emission level for an electronic component over its lifecycle. This value underscores the product's environmental impact, primarily driven by energy consumption patterns rather than material intensity. The lifecycle stage contributions reveal a highly skewed distribution, with the use phase accounting for 218.167 kg CO₂-Eq or 92.4% of total emissions, while raw material extraction, production, distribution, and end-of-life stages collectively contribute less than 8%. This characteristic highlights the dominance of operational electricity use in the product's environmental profile.

Major emission sources are concentrated in the use stage, indicating that electricity consumption during operation is the primary driver. Secondary sources include raw material processing at 3.2% and production activities at 2.1%, reflecting upstream energy and material inputs. The minimal contributions from distribution and end-of-life phases suggest efficient logistics and limited disposal impacts, but the overall profile emphasizes the need for targeted interventions in energy-intensive operational phases to achieve meaningful reductions.

2. Statement of Assumptions and Limitations

Key assumptions in this study include the use of average European electricity grid mixes for the use phase modeling and standard product lifetimes based on manufacturer specifications. It is also assumed that end-of-life scenarios follow typical recycling rates without accounting for regional variations in waste management infrastructure. These assumptions facilitate standardized calculations but may not fully capture site-specific conditions or future technological shifts in energy systems.

Limitations of data sources and quality stem from reliance on secondary databases such as Ecoinvent, which may contain aggregated or outdated emission factors for specialized LED components. Primary data gaps exist for proprietary production processes, leading to potential underestimation of variability in material compositions or manufacturing efficiencies. Data quality is further constrained by allocation methods for multi-output processes, introducing uncertainty in impact attribution.

System boundary restrictions encompass a cradle-to-grave approach limited to the defined lifecycle stages, excluding indirect effects such as infrastructure development or consumer behavior variations. The boundaries do not extend to upstream supply chain tiers beyond tier-one suppliers or downstream impacts from product reuse, potentially omitting broader environmental interactions.

3. Suggestions for Improvement

- Target the use phase by developing higher-efficiency LED drivers and integrating smart controls to reduce electricity demand by up to 30%, directly addressing the 92.4% emission share.

- Shift to renewable energy sourcing for operational electricity in end-user applications, such as through on-site solar integration or certified green tariffs, to lower the carbon intensity of the dominant lifecycle stage.
- Optimize raw material selection by increasing recycled content in semiconductor substrates and housings to cut the 3.2% contribution from extraction processes.
- Implement closed-loop recycling programs at end-of-life to enhance material recovery rates beyond current 1.2% impacts, supported by design-for-disassembly features.
- Improve data quality by conducting primary data collection from suppliers for production and use phases, supplemented by sensitivity analyses to refine uncertainty in electricity mix assumptions.